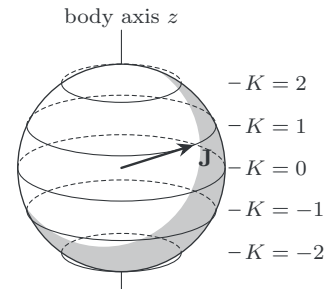
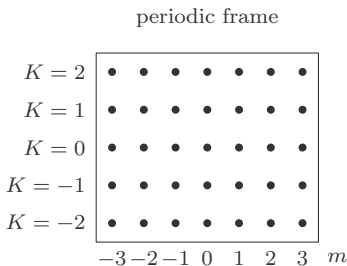


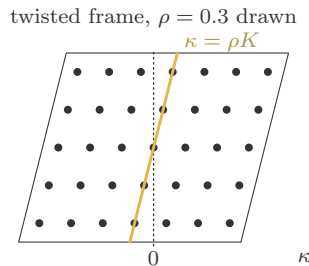
one state, two bases: $|r, \tau, Q\rangle_0$ and $|JMK, m, p\rangle$, the chart density absorbed in $\tilde{\psi} = \mathcal{J}^{1/2}\psi$; no energies



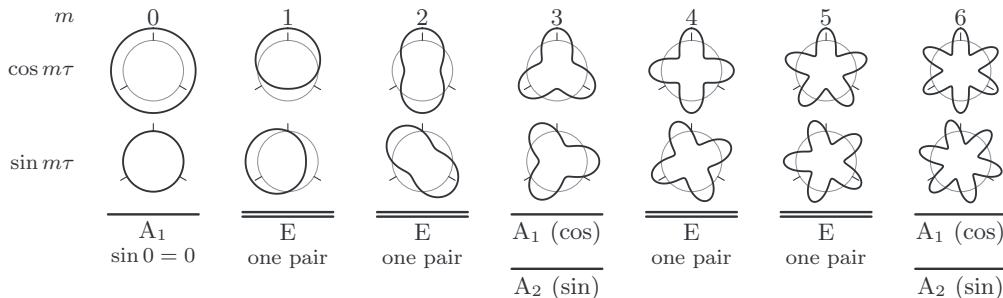
$J = 2$: $\mathbf{J}$ of length $\sqrt{J(J+1)}$
on a latitude ring at height
 K , one row at right per K



a turn of the body frame at the seam, $(r, \tau + 2\pi, Q) \sim (rR_z(-2\pi\rho), \tau, Q)$, shears the torsional labels row by row: $\kappa = m + \rho K$



periodic torsion modes on the circle of figure 9, the three version marks at $\tau = 0, 2\pi/3, 4\pi/3$; $U_t f_m = e^{-2\pi i m/3} f_m$, $U_b f_m = f_{-m}$



a single rule for a one-dimensional species, a double rule for E; the modes with $3 \mid m$ are the ones invariant under the third-turn t